ECON 131: Midterm Preparation

Reference: Bekaert & Hodrick, International Financial Management (Chapters 2–6 & 8)

1. Chapters 2 & 3: Spot Markets, Spreads, and Forward Risk

1. Deutsche Bank quotes the following bilateral bid–ask spreads:

$$S(\text{USD/EUR}) = \$1.3005 - 1.3007/€$$
$$S(\text{JPY/USD}) = ¥104.30 - 104.40/\$$$

   Determine Deutsche Bank's direct asking price and direct bid price of Japanese yen per euro (¥/€).

2. Assume that you are the financial manager of Zachy's, a wine store in Scarsdale, New York, that imports wine from France. You have just contracted to import some Chateau Margaux wine, and your invoice is for €4,000,000. You have agreed to pay this number of euros when you have received the wine and determined that it is in good condition. Payment of the euros and delivery of the wine are scheduled for 90 days in the future. The following market data are available:

   - Today's spot rate: $S = \$1.10/€$
   - Today's 90-day forward rate: $F_{90} = \$1.08/€$
   - 90-day dollar interest rate = 6% p.a.
   - 90-day euro interest rate = 13.519% p.a.

   What is the source of your transaction exchange risk, and how much could you lose if the position is left completely unhedged? How can you eliminate this risk in a forward hedge and money market hedge? Which among the two hedging strategies would you prefer?

2. Chapter 4: Balance of Payments Accounting

1. Consider an Indonesian resident who in previous years invested in Japanese government bonds. Each year, the Indonesian receives ¥500,000 of coupon payments from her Japanese bonds. Suppose that these payments are paid to her Tokyo bank, where she keeps a yen-denominated bank account. What are the credit and debit items on the Indonesian balance of payments?

2. What are the effects on the French balance of payments of the following set of transactions? Les Fleurs de France, the French subsidiary of a British company, The Flowers of Britain, has just received €4.4 million of additional investment from its British parent. Part of the investment is a €0.9 million computer system that was shipped from Britain directly. The €3.5 million remainder was financed by the parent by issuing euro-denominated Eurobonds to investors outside of France. Les Fleurs de France is holding these euros in its Paris bank account.

3. Chapter 6: Covered Interest Parity (CIP)

1. Suppose the 5-year annualized interest rate on a dollar-denominated pure discount bond is 4.5% p.a., and the interest rate on a comparable euro-denominated pure discount bond is 7.5% p.a. (with annual compounding). If the current spot rate is \$1.08/€, what 5-year forward exchange rate $F_5(\$/€)$ eliminates covered interest arbitrage?

2. A portfolio manager at Deutsche Bank has €10,000,000 to invest for 180 days with no exchange rate risk. Market data (360-day year, $t = 180/360 = 0.5$):

   - 180-day Swiss franc (CHF) interest rate: 8.0% p.a.
   - 180-day Euro (EUR) interest rate: 10.0% p.a.
   - Spot exchange rate: €1.1960/CHF
   - 180-day forward exchange rate: €1.2024/CHF

   Which deposit alternative yields the higher euro return after 180 days? If these prices prevailed in the market, what market forces would re-establish parity?

3. If the 30-day yen interest rate is 3.0% p.a. and the 30-day euro interest rate is 5.0% p.a. (using a 360-day year convention), what is the annualized percentage forward premium or discount on the yen (¥) relative to the euro (€)?

4. Suppose the spot rate is CHF 1.4706/$ and the 180-day forward rate is CHF 1.4295/$. If the 180-day annualized dollar interest rate is 7.0% p.a. (180/360 day count), what annualized 180-day Swiss franc interest rate i_{CHF} prevents covered interest arbitrage?

4. Worked Solutions

4.1. Chapters 2 & 3: Spot Markets, Spreads, and Forward Risk

Question 1: Cross-rate bid and ask

Given:

$$S(\text{USD/EUR}) = 1.3005 - 1.3007, \qquad S(\text{JPY/USD}) = 104.30 - 104.40.$$

The cross-rate bid is the product of the two bid rates:

$$\begin{aligned} S_{\text{bid}}(\text{JPY/EUR}) &= 1.3005 \times 104.30 \\ &= 135.64215 \text{ JPY/EUR}. \end{aligned}$$

The cross-rate ask is the product of the two ask rates:

$$\begin{aligned} S_{\text{ask}}(\text{JPY/EUR}) &= 1.3007 \times 104.40 \\ &= 135.79308 \text{ JPY/EUR}. \end{aligned}$$

Therefore,

$$\boxed{S(\text{JPY/EUR}) = 135.6422 - 135.7931 \text{ JPY/EUR}}$$

The bid is lower than the ask because the bank buys currency at a lower price than it sells it.

Question 2: Zachy's transaction exchange risk

(a) Source of risk. Zachy's must pay €4,000,000 in 90 days. Since the invoice is denominated in euros while the firm's domestic currency is the dollar, the dollar cost of the payment is uncertain.

Zachy's is exposed to transaction exchange risk: if the euro appreciates against the dollar, the firm must spend more dollars to purchase the euros needed for payment.

At today's spot rate, the dollar equivalent is

$$€4,000,000 \times \$1.10/€ = \$4,400,000.$$

(b) Potential unhedged loss. The actual loss depends on the future spot rate. For example, if the euro appreciates to \$1.20/€, the payment costs

$$€4,000,000 \times \$1.20/€ = \$4,800,000.$$

The additional cost relative to today's spot-rate valuation is

$$\$4,800,000 - \$4,400,000 = \boxed{\$400,000}.$$

There is no finite maximum loss implied by the information given, because the future exchange rate is unknown.

(c) Forward hedge. Zachy's can enter a 90-day forward contract to buy €4,000,000 at \$1.08/€.

The dollar cost is locked in at

$$€4,000,000 \times \$1.08/€ = \boxed{\$4,320,000}.$$

This eliminates the exchange-rate uncertainty associated with the payment, assuming the forward contract matches the amount and maturity of the invoice.

(d) Money market hedge. A money market hedge replicates a forward hedge by borrowing and lending in the two currencies.

First, calculate the present value of the euro payment. Using the annualized rates and a 90/360 day-count convention:

$$PV_{€} = \frac{€4,000,000}{1 + 0.13519(90/360)}.$$

Thus,

$$PV_{€} \approx €3,869,393.$$

Zachy's should borrow this amount of euros today, convert the proceeds into dollars at the spot rate, and invest or retain the dollars as appropriate.

The dollar amount needed today is

$$€3,869,393 \times \$1.10/€ \approx \$4,256,332.$$

The dollar repayment after 90 days is

$$\$4,256,332\,[1 + 0.06(90/360)] \approx \boxed{\$4,320,000}.$$

The euro loan grows to approximately €4,000,000, which can be used to pay the invoice on its due date.

(e) Comparison of hedging strategies.

Strategy	**Dollar cost in 90 days**
Forward hedge	$4,320,000
Money market hedge	Approximately $4,320,000

The two strategies produce approximately the same cost because the forward rate and interest rates are consistent with covered interest parity, subject to rounding.

4.2. Chapter 4: Balance of Payments Accounting

Question 1: Indonesian resident's coupon payments

The coupon payment is investment income received from Japan. It is recorded as a credit in the Indonesian current account, under primary income.

Because the payment remains in the resident's Tokyo bank account, the resident acquires an additional financial asset abroad. This is recorded as a debit in the Indonesian financial account.

BOP account	**Entry**	**Amount**
Current account: primary income	Credit	¥500,000
Financial account: foreign bank deposits	Debit	¥500,000

The two entries offset each other. The resident earns income from abroad and retains the proceeds as a foreign bank deposit.

Question 2: Investment in the French subsidiary

The French subsidiary receives €4.4 million in additional investment from its British parent.

The computer system shipped from Britain is an import of goods by France:

$$\text{Goods import} = €0.9 \text{ million} : \textit{debit(-) in current account}$$

$$\text{Foreign ownership of French assets} = €0.9 \text{ million} : \textit{credit (+) in capital account}$$

The British parent acquired an equity claim on the French subsidiary, so foreigners acquired a French asset. However, the euros did not materialize in Paris out of nowhere. They were paid out of investors' bank accounts outside France and settled into the French banking system. When the Paris bank credited Les Fleurs' account, it acquired €3.5M of claims on banks abroad (its own euro balances at foreign correspondent banks). That rise in French-owned claims on foreigners is the debit.

$$\text{Foreign purchases of French assets} = €3.5 \text{ million.} : \textit{credit (+) in capital account}$$

$$\text{French purchases of foreign assets} = €3.5 \text{ million.} : \textit{debit (-) in capital account}$$

4.3. Chapter 6: Covered Interest Parity (CIP)

The covered interest parity relationship requires that the return on a domestic deposit equals the exchange-rate-hedged return on a foreign deposit.

When the exchange rate is quoted as dollars per euro,

$$F_t(\$/€) = S(\$/€)\frac{(1+i_{\$})^t}{(1+i_{€})^t}$$

for annually compounded rates over t years.

For short-term rates quoted on a 360-day basis, the interest factors are typically written as $1 + it$.

Question 1: Five-year forward exchange rate

Given:

$$S = \$1.08/€, \quad i_{\$} = 0.045, \quad i_{€} = 0.075, \quad t = 5.$$

By covered interest parity,

$$F_5 = 1.08\frac{(1.045)^5}{(1.075)^5}.$$

Therefore,

$$\boxed{F_5 \approx \$0.9375/€}.$$

The euro has a forward discount in the dollar-per-euro quotation because the euro interest rate is higher than the dollar interest rate.

Question 2: Swiss franc versus euro deposits

Given:

$$\begin{aligned} \text{Initial funds} &= €10,000,000, \\ i_{\text{CHF}} &= 0.08, \\ i_{€} &= 0.10, \\ S &= €1.1960/\text{CHF}, \\ F &= €1.2024/\text{CHF}, \\ t &= 0.5. \end{aligned}$$

Alternative A: Euro deposit. The euro investment grows to

$$€10,000,000(1 + 0.10 \times 0.5) = \boxed{€10,500,000}.$$

Alternative B: Swiss franc deposit with a forward hedge. First, convert the initial euros into Swiss francs:

$$\text{CHF}_0 = \frac{€10,000,000}{1.1960} \approx \text{CHF } 8,361,204.01.$$

Next, earn interest for 180 days:

$$\begin{aligned} \text{CHF}_{180} &= 8,361,204.01(1 + 0.08 \times 0.5) \\ &\approx \text{CHF } 8,695,652.17. \end{aligned}$$

Convert the maturity value into euros at the forward rate:

$$\begin{aligned} \text{EUR}_{180} &= 8,695,652.17 \times 1.2024 \\ &\approx \boxed{€10,456,052}. \end{aligned}$$

Comparison.

Investment alternative	**Final euro value**
Euro deposit	€10,500,000
Hedged Swiss franc deposit	€10,456,052
Difference	€43,948

The euro deposit yields approximately €43,948 more.

Market forces restoring parity. Investors could borrow Swiss francs, convert them into euros at the spot rate, invest the euros at the higher euro interest rate, and sell the future euro proceeds forward to hedge the exchange risk.

These trades increase spot demand for euros and forward supply of euros. Borrowing and lending also put pressure on interest rates. These adjustments tend to eliminate the covered return differential and restore CIP.

Question 3: Annualized yen forward premium or discount

Given:

$$i_{¥} = 0.03, \quad i_{€} = 0.05, \quad t = \frac{30}{360} = \frac{1}{12}.$$

For an exchange rate quoted in euros per yen, CIP implies

$$\frac{F - S}{S} = \frac{1 + i_{€} t}{1 + i_{¥} t} - 1.$$

Substituting:

$$\begin{aligned} \frac{F - S}{S} &= \frac{1 + 0.05/12}{1 + 0.03/12} - 1 \\ &\approx 0.0016625. \end{aligned}$$

Annualizing using the 360-day convention:

$$\begin{aligned} \text{Annualized forward premium} &= 0.0016625 \left(\frac{360}{30} \right) 100 \\ &\approx \boxed{2.00\% \text{ p.a.}} \end{aligned}$$

Thus, the yen has an annualized forward premium of approximately 2.00% when the exchange rate is quoted in euros per yen. Equivalently, in the inverse yen-per-euro quotation, the yen has a forward discount.

Question 4: Swiss franc interest rate

Given:

$$\begin{aligned} S &= \text{CHF } 1.4706/\$, \\ F &= \text{CHF } 1.4295/\$, \\ i_{\$} &= 0.07, \\ t &= 0.5. \end{aligned}$$

With the exchange rate quoted in Swiss francs per dollar, covered interest parity gives

$$F = S \frac{1 + i_{\text{CHF}} t}{1 + i_{\$} t}.$$

Rearranging:

$$i_{\text{CHF}} = \frac{\frac{F}{S}(1 + i_{\$} t) - 1}{t}.$$

Substituting:

$$i_{\text{CHF}} = \frac{\left(\frac{1.4295}{1.4706}\right)(1 + 0.07 \times 0.5) - 1}{0.5}.$$

Therefore,

$$\boxed{i_{\text{CHF}} \approx 1.21\% \text{ p.a.}}$$

The Swiss franc interest rate must be substantially lower than the dollar interest rate to offset the dollar's forward discount in the CHF-per-dollar quotation.